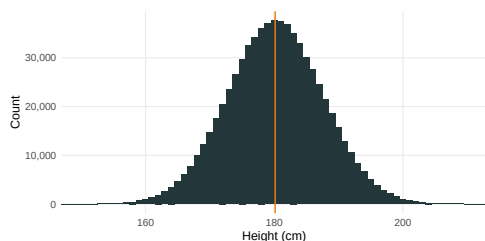


Week 1, part 2: notes on the figures

Intermediate Quantitative Methods, 2026–27.

All figures are simulated with `figures/inference_figures.R` (seed 20261002), so every number below comes from that script.



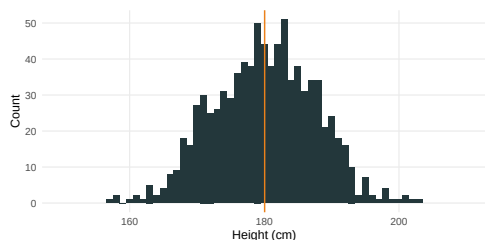
Population: 700,000 simulated heights

On screen

- 700,000 draws from $N(180.15, 7.4)$, the Dutch height example from the 2024 deck
- Orange line: the population mean $\mu = 180.15$

Say

- This is the view we never get. We only have it because we simulated it.
- That is the point of simulation in this course: set the truth yourself, then check whether a method recovers it.
- μ and σ are fixed, unknown numbers. Everything after this slide is about guessing them.



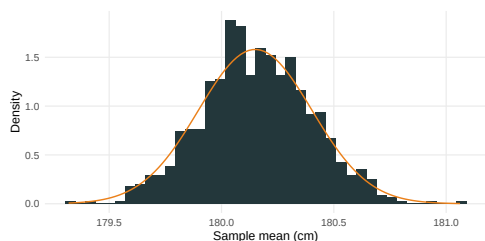
One random sample, $n = 859$

On screen

- One sample of 859 from that population
- Orange line: the sample mean, $\bar{x} = 180.05$, with $s = 7.52$

Say

- This is everything a real researcher has.
- The shape looks like the population, just noisier.
- $\bar{x}$ misses μ by 0.1 cm. From this one sample we cannot tell whether the miss is 0.1 cm or 3 cm. That is the inference problem.



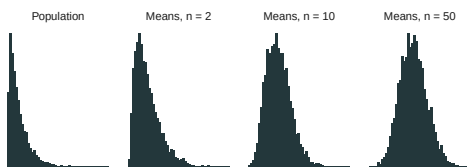
Means of 1,000 samples, $n = 859$

On screen

- Histogram: means of 1,000 separate samples of 859
- Orange curve: normal density with mean μ and SD $\sigma/\sqrt{n} = 0.25$

Say

- Point to the x-axis: the whole range is about 179.5 to 181, against 150 to 210 for the population.
- The SD of these 1,000 means is 0.24, close to the theoretical 0.25. That SD *is* the standard error.
- In practice we never draw 1,000 samples. Theory tells us the shape and spread, which is what lets us reason from a single sample.



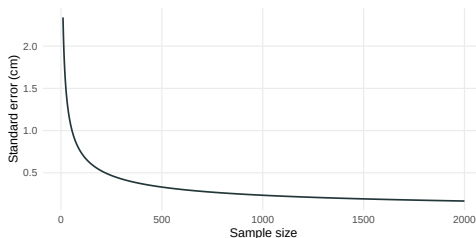
Skewed population, means of 5,000 samples

On screen

- Left: a heavily right-skewed (exponential) population
- Then: means of 5,000 samples each, at $n = 2$, 10 and 50
- Axes are free and unlabeled: only the shape matters

Say

- At $n = 2$ the means are still skewed. By $n = 10$ they are close to symmetric. At $n = 50$ they are close to a bell curve.
- This is why normal-based tests work for means of income, counts and other non-normal variables.
- The caveat is independence. Link to Lab 2: 30,000 ESS respondents clustered in 24 countries carry much less information than 30,000 independent draws.



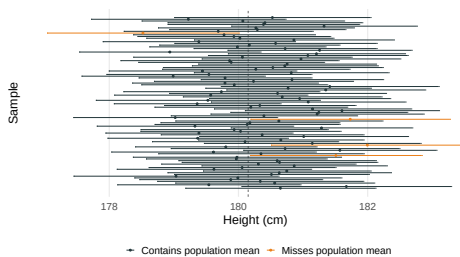
Standard error by sample size, $\sigma = 7.4$

On screen

- $SE = 7.4/\sqrt{n}$ for n from 10 to 2,000

Say

- Diminishing returns. The SE is 2.34 at $n = 10$, 0.74 at $n = 100$ and 0.23 at $n = 1,000$.
- Four times the data halves the SE. Going from 100 to 1,000 buys much less than going from 10 to 100.
- This matters for survey budgets and for power later in the lecture.



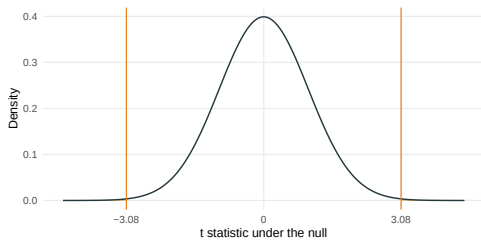
100 samples of 100, 100 95% intervals

On screen

- Each row: one sample of $n = 100$ and its 95% t -interval
- Dashed line: μ
- Orange: intervals that miss μ . In this draw 4 of 100 miss (5 expected).

Say

- Each line is a different researcher with a different sample.
- The 95% describes the procedure: about 95 of every 100 lines cross the dashed line.
- No researcher knows whether their own line is orange. You can ask the room which line is theirs.



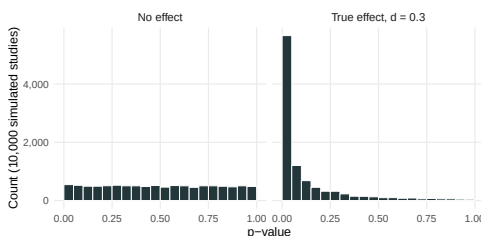
Null distribution, $t = \pm 3.08$

On screen

- t distribution with 2,624 degrees of freedom, which is effectively standard normal
- Orange lines at ± 3.08 , the North/South Italy age example

Say

- The p-value is the area in the two tails beyond the orange lines. Here it is 0.002, so small you can barely see it.
- The test is two-sided: we would have been just as surprised by a difference of -2.44 .
- Everything here assumes H_0 is true. The p-value says nothing about how likely H_0 is.



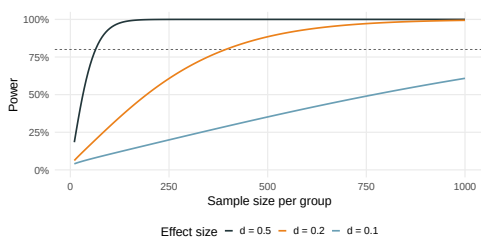
p-values from 10,000 simulated studies

On screen

- 10,000 simulated two-group studies with $n = 100$ per group
- Left: no true effect. Right: a true effect of $d = 0.3$.
- Bars are 0.05 wide, so the leftmost bar is $p < 0.05$.

Say

- Left panel: the histogram is flat. Under the null, $p = 0.04$ is no rarer than $p = 0.74$. About 5% fall below 0.05 (5.5% in this draw), which is α .
- Right panel: p-values pile up near zero. 57% are below 0.05, which is the power of this design (theory: 56%).
- Even with a real, moderate effect, 43% of these studies “fail.” A p-value is a random variable, not a property of the effect.



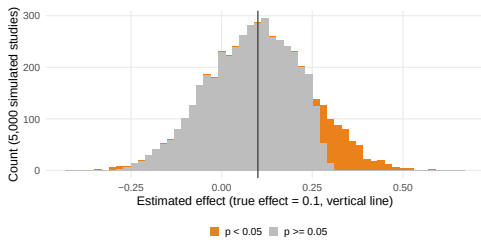
Power by sample size, two groups, $\alpha = 0.05$

On screen

- Power of a two-sample t -test by n per group, for $d = 0.5$, 0.2 and 0.1
- Dashed line: 80%

Say

- n per group needed for 80% power: 64 for $d = 0.5$, 394 for $d = 0.2$ and 1,571 for $d = 0.1$.
- Check what effect sizes are typical in your literature before assuming a medium effect.
- A survey experiment with 500 per arm is fine for $d = 0.2$ and badly underpowered for $d = 0.1$.



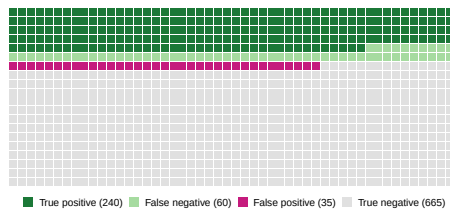
5,000 studies, true effect 0.1, $n = 100$ per group

On screen

- Estimated effects from 5,000 simulated studies with a true $d = 0.1$
- Orange: significant at 0.05. Grey: not significant.
- Vertical line: the true effect
- Power is about 11%, roughly the political science median

Say

- With $SE \approx 0.14$, an estimate has to exceed about 0.28 to be significant, nearly three times the truth.
- Significant estimates average 0.34 in absolute value, about 3 times the true effect (Type M). About 3% of them have the wrong sign (Type S).
- If only the orange studies are published, the literature is exaggerated by construction, even with no misconduct.



1,000 hypotheses: 30% true, 80% power, $\alpha = 0.05$

On screen

- 1,000 hypotheses: 300 true, 700 false
- Dark green: true positives (240). Light green: false negatives (60).
- Magenta: false positives (35). Grey: true negatives (665).

Say

- Carried over from the 2024 deck (see Colquhoun 2014 in the optional readings).
- Of 275 significant results, 35 are false: 13%.
- Next slide, at 10% power: 30 true positives against 35 false, so 54% of “findings” are false.
- The prior also matters. With 10% of hypotheses true and 80% power: 80 true against 45 false, so 36% are false.